Empirical Boundaries of Feature-Gated Consensus Clustering: Permutation-Null Calibration and Hurdle Recovery on High-Dimensional Manifolds

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Abstract

Purpose: High-dimensional unsupervised clustering ($\boldsymbol{D \gg N}$) is fundamentally challenged by the curse of dimensionality, non-informative distractor features, and technical artifacts such as dropout zeros. While heuristic feature gating and consensus ensembles are widely employed, their rigorous statistical necessity, empirical boundaries, and failure modes remain poorly characterized.

Methods: Following an unalterable pre-registered confirmatory protocol (tag `freeze-v1`, Rule R2), we evaluate Permutation-Null Calibrated Consensus Clustering (NullConsensus) and Two-Part Hurdle Gating (HurdleGate) against 11 established and contemporary baselines across 27 fresh benchmark cohorts spanning 6,880 independent model fits. The evaluation encompasses 15 high-dimensional oncology microarrays ($\boldsymbol{D/N > 20}$), 4 zero-inflated single-cell RNA-seq count cohorts, 5 multi-class benchmarks, and 3 physical controls.

Results: We report three decisive findings: (1) On continuous oncology microarrays, the primary confirmatory hypothesis that permutation-null gating universally surpasses matched uniform random subspaces is falsified ($\boldsymbol{\Delta}$ARI $= \boldsymbol{-0.0433}$, $\boldsymbol{p = 0.2293}$). Uniform random gating achieved Rank #3 (Mean ARI = 0.263) while null screening achieved Rank #8 (0.2113), indicating that unimodal reference nulls behave over-conservatively on small sample sizes ($\boldsymbol{N < 100}$). (2) Across the full 27-dataset omnibus benchmark, NullConsensus demonstrates a statistically significant improvement over Paper 1's GAC pipeline (icc_v2) with $\boldsymbol{\Delta}$ARI $= \boldsymbol{+0.0542}$ (95% CI: $\boldsymbol{[+0.0137, +0.0989]}$, $\boldsymbol{p = 1.1200e-02}$), and over CoHiRF ($\boldsymbol{\Delta}$ARI $= \boldsymbol{+0.0689}$, $\boldsymbol{p = 5.5203e-02}$).

(3) On zero-inflated scRNA-seq manifolds, conventional bimodality screening collapses to near-zero (ARI $\approx \mathbf{0.04}$) due to dropout point masses. HurdleGate isolates Bernoulli dropout dispersion from continuous support multimodality, driving an extraordinary performance recovery ($\mathbf{\Delta}$ARI $= \mathbf{+0.3832}$, 95% CI: $\mathbf{[+0.3157, +0.4596]}$, $\boldsymbol{p = 3.57e - 08}$), validating Hypothesis H2.
Conclusion: Feature gating provides no universal advantage over uniform random subspaces in continuous Gaussian regimes, but is essential for isolating biological manifold structure in zero-inflated count geometries. We synthesize these empirical boundaries into a formal regime taxonomy for high-dimensional clustering.

Keywords: Consensus Clustering, Feature Gating, High-Dimensional Data, Permutation Null, Zero-Inflation, Single-Cell RNA-seq, Confirmatory Benchmark

1 Introduction

Unsupervised cluster analysis on high-dimensional biological and physical manifolds where the feature dimension D substantially exceeds the sample size N ($D \gg N$) is central to modern computational biology, functional genomics, and image analysis [1, 2]. In oncology microarrays and single-cell transcriptomics, D routinely ranges from 10^3 to 10^5, while sample cohorts frequently contain fewer than 100 patient specimens [2, 3]. Under these conditions, the concentration of distances and the presence of thousands of isotropic distractor features degrade the metric space of classical partitioning solvers such as K-Means and Ward's hierarchical agglomerative clustering [4, 5].

To mitigate distractor degradation, two primary paradigms dominate the literature:

1. **Random Subspace Consensus Ensembles:** Generating M perturbed partitions over randomly sampled feature subsets and accumulating co-association evidence into a consensus matrix $\mathbf{S} \in [0, 1]^{N \times N}$ [1, 5, 6].
2. **Marginal Feature Gating:** Pre-filtering features using unsupervised distributional statistics, such as Sarle's Bimodality Coefficient (BC), to select a candidate pool $\mathcal{F}_{\text{active}} \subset \{1, \ldots, D\}$ prior to consensus assembly [7, 8].

Despite the widespread adoption of feature gating, its fundamental empirical boundaries remain largely unexplored. In particular, existing literature rarely answers two critical questions:

- *Does unsupervised feature screening provide a statistically significant advantage over matched uniform random subspace selection, or is the observed gain an artifact of ensemble averaging?*
- *How do technical artifacts, specifically extreme dropout zeros in single-cell transcriptomics, interact with marginal multimodality screening?*

In this work, we conduct a pre-registered confirmatory benchmark across 27 sealed datasets spanning 6880 independent fits. We introduce NullConsensus, an anchor

consensus architecture calibrated against finite-sample unimodal permutation nulls (NullGate), and HurdleGate, a two-part decomposition separating Bernoulli dropout point masses from continuous positive-support multimodality.

Our empirical findings falsify the hypothesis that null-calibrated gating universally outperforms random subspaces on continuous microarray data ($p_{\text{holm}} = 0.1437$), while decisively confirming its indispensable role on zero-inflated single-cell manifolds ($\Delta\text{ARI} = +0.3832$, $p = 3.57e-08$).

2 Theoretical Formulations & Algorithms

2.1 Bipartite Landmark Anchor Consensus

Classical consensus clustering constructs an $N \times N$ dense co-association matrix requiring $\mathcal{O}(N^2)$ memory and $\mathcal{O}(N^3)$ spectral decomposition time [1, 6]. To scale to large single-cell cohorts ($N > 1{,}000$), we utilize a bipartite landmark graph [8, 9].

Let $\mathbf{X} \in \mathbb{R}^{N \times D}$ denote the centered data matrix. We select $L \ll N$ landmark anchor points $\mathcal{A} \subset \{1, \dots, N\}$ (with default $L = 40$). Across $M = 25$ base clusterings generated on subspace projections, we construct a rectangular co-association matrix $\mathbf{B} \in [0, 1]^{N \times L}$:

$$B_{il} = \frac{1}{M} \sum_{m=1}^{M} \mathbf{1} \left\{ \pi_m(i) = \pi_m(l) \right\}, \tag{1}$$

where $\pi_m(i) \in \{1, \dots, K\}$ denotes the cluster assignment of sample i in partition m.

The bipartite graph Laplacian is solved via the Transfer-Cut formulation [9]:

$$\mathbf{W}_{\text{anchor}} = \mathbf{B}_{\text{norm}}^T \mathbf{B}_{\text{norm}} \in \mathbb{R}^{L \times L}, \tag{2}$$

where $\mathbf{B}_{\text{norm}} = \mathbf{D}_r^{-1} \mathbf{B}$ is row-stochastic. The spectral embedding $\mathbf{U} \in \mathbb{R}^{N \times K}$ is obtained in linear $\mathcal{O}(NLK)$ time via the generalized eigenvectors of the compact $L \times L$ anchor kernel:

$$\mathbf{U} = \mathbf{B}_{\text{norm}} \mathbf{D}_c^{-1/2} \mathbf{V}_K \mathbf{\Sigma}_K^{-1/2}, \tag{3}$$

where $\mathbf{V}_K$ and $\mathbf{\Sigma}_K$ are the top eigenvectors and eigenvalues of $\mathbf{D}_c^{-1/2} \mathbf{W}_{\text{anchor}} \mathbf{D}_c^{-1/2}$, completely bypassing $\mathcal{O}(N^2)$ storage.

2.2 Permutation-Null Calibrated Gating (NullGate)

Standard feature gating filters features via Sarle's Bimodality Coefficient:

$$\text{BC}_d = \frac{\gamma_d^2 + 1}{\kappa_d + 3}, \tag{4}$$

where γ_d is sample skewness and κ_d is excess kurtosis. However, in finite samples, sample moments exhibit high stochastic variance.

NullGate evaluates empirical BC_d against a finite-sample unimodal reference distribution:

$$z_d = \frac{\text{BC}_d - \mu_{\text{null}}(N)}{\sigma_{\text{null}}(N)}, \tag{5}$$

Table 1 Summary of the 27 sealed benchmark datasets evaluated in the confirmatory protocol (Rule R2).

Domain	Dataset Name	Reference / Source	N	D	K	D/N	Majority %
Oncology Microarray ($D/N > 20$)							
	armstrong2002	High-D cancer benchmark for ML	72	2194	3	30.5	38.9%
	bhattacharjee2001	Multi-class lung adenocarcinom	203	1543	5	7.6	68.5%
	chowdary2006	Moderate D/N breast vs colon t	104	182	2	1.8	59.6%
	dyrskjot2003	Small-N high-D bladder tumor g	40	1203	3	30.1	50.0%
	gordon2002	Malignant pleural mesothelioma	181	1626	2	9.0	82.9%
	pomeroy2002	Ultra-high D/N CNS embryonal t	42	1379	5	32.8	23.8%
	ramaswamy2001	Large-K (K=14) ultra-high-dime	190	1363	14	7.2	15.8%
	shipp2002	Diffuse large B-cell lymphoma	77	798	2	10.4	75.3%
	singh2002	Extreme D/N prostate tumor vs	102	339	2	3.3	51.0%
	su2001	Multi-tissue human carcinoma p	174	1571	10	9.0	16.1%
	west2001	Ultra-small N (N=49) breast ca	49	1198	2	24.4	51.0%
	yeoh2002	Six-subtype acute lymphoblasti	248	2526	6	10.2	31.9%
	alizadeh2000	B-cell lymphoma molecular clas	42	1095	2	26.1	50.0%
	nutt2003	Extreme D/N (252.5) high-grade	50	1377	4	27.5	30.0%
	lapointe2004	Prostate cancer progression su	69	1625	3	23.6	56.5%
Single-Cell RNA-seq (Zero-Inflated)							
	zheng_b_cells	Evaluation of HurdleGate on hi	976	13714	3	14.1	49.2%
	zheng_t_cells	Evaluation of zero-inflated im	1000	13714	4	13.7	66.2%
	klein_embryonic	Four-stage developmental diffe	1000	3451	4	3.5	34.2%
	baron_pancreas	Eight pancreatic endocrine/exo	1147	2000	8	1.7	46.4%
Multi-Class Benchmarks ($K \geq 10$)							
	coil20	Large-K benchmark (K=20) for t	2000	240	10	0.1	10.0%
	umist_faces	Large-K facial identity cluste	575	644	20	1.1	8.3%
	texture16	Brodatz texture classification	4898	11	7	0.0	44.9%
	letter_recognition	Low-D large-K control (K=26) c	2000	16	26	0.0	4.8%
	news20_subset	Fresh 15-category partition of	1500	2000	15	1.3	7.7%
Physical / Biomedical Controls							
	parkinsons	Low-dimensional clinical vocal	195	22	2	0.1	75.4%
	wdbc	Classical low-D pathological c	569	30	2	0.1	62.7%
	ionosphere	Non-omics physical radar contr	351	34	2	0.1	64.1%

where $\mu_{\text{null}}(N)$ and $\sigma_{\text{null}}(N)$ are computed via Monte Carlo draws of size N from $\mathcal{N}(0,1)$. Features are sampled with softmax temperature T:

$$p_d = \frac{\exp(z_d^+/T)}{\sum_{j\in\mathcal{F}_{\text{pool}}}\exp(z_j^+/T)}, \quad z_d^+ = \max(0, z_d). \tag{6}$$

2.3 Two-Part Hurdle Gating (HurdleGate)

In single-cell RNA-sequencing (scRNA-seq), reverse transcription inefficiencies create technical dropouts, yielding point masses at zero for 60%–90% of entries. This point mass artificially inflates sample skewness and kurtosis, driving BC $\to$ 1.0 even for purely uninformative noise genes.

HurdleGate resolves this pathology via a two-part decomposition:

$$s_d = \alpha \cdot \text{Entropy}(p_{0,d}) + (1-\alpha)\cdot \text{BC}_d(\mathbf{x}_d^+), \tag{7}$$

where $p_{0,d} = \frac{1}{N}\sum_{i=1}^{N}\mathbf{1}\{x_{id}=0\}$ is the dropout fraction, and $\text{BC}_d(\mathbf{x}_d^+)$ evaluates multimodality exclusively on expressing cells $\mathbf{x}_d^+ = \{x_{id} \mid x_{id} > 0\}$. This decouples dropout-driven binary entropy from true continuous subtype expression.

Table 2 Empirical Adjusted Rand Index (mean ± std over 30 independent seeds) across the 15 high-dimensional cancer cohorts ($D/N > 20$). Bold indicates the highest mean performance per cohort.

Dataset	K-Means	Ward Hier.	Spectral	Std. Consensus	Core-Halo	FCAG	Sparse KM	CoHiRF	M3C	ICC-v2 (GAC)	Random Gate	NullConsensus
armstrong2002	0.443	**0.821**	0.422	0.632	0.633	0.468	0.552	0.311	0.598	0.136	0.554	0.313
bhattacharjee2001	0.326	0.378	0.469	0.313	0.092	0.309	0.247	0.384	0.357	0.275	0.274	**0.637**
chowdary2006	0.066	0.066	**0.924**	0.060	0.013	0.813	0.057	0.053	0.066	0.022	0.172	0.052
dyrskjot2003	0.272	0.107	0.336	0.238	0.396	**0.411**	0.232	0.020	0.192	0.121	0.305	0.129
gordon2002	**0.973**	0.918	0.890	0.952	-0.064	0.449	0.821	0.756	0.971	0.295	0.825	0.624
pomeroy2002	0.189	0.313	0.333	0.075	**0.414**	0.263	0.098	0.041	0.062	0.179	0.377	0.322
ramaswamy2001	0.035	0.129	**0.268**	0.026	0.116	0.016	0.017	0.002	0.027	0.020	0.103	0.035
shipp2002	-0.054	**0.052**	-0.098	-0.048	-0.001	-0.069	-0.062	-0.012	-0.051	0.030	-0.031	-0.003
singh2002	0.041	0.026	0.033	0.040	0.000	0.042	0.041	0.041	0.041	0.040	0.046	**0.049**
su2001	0.565	0.645	**0.688**	0.665	0.559	0.478	0.378	0.113	0.668	0.436	0.546	0.405
west2001	-0.002	-0.002	0.291	-0.001	0.247	0.001	0.019	-0.001	-0.001	0.259	0.101	**0.309**
yeoh2002	-0.002	0.013	**0.167**	0.002	-0.001	0.001	0.000	-0.001	0.001	0.056	0.003	0.119
alizadeh2000	0.000	0.000	**0.090**	0.000	0.000	0.000	0.000	0.000	0.000	-0.011	0.000	-0.017
nutt2003	0.389	0.406	0.146	**0.419**	0.160	0.290	0.326	0.081	0.406	0.154	0.380	0.105
lapointe2004	**0.176**	0.157	0.024	0.167	0.068	0.174	0.134	0.133	0.158	0.081	0.165	0.091
Mean	0.228	0.269	**0.332**	0.236	0.175	0.243	0.191	0.128	0.233	0.140	0.255	0.211
Avg Rank	#6	#5	#5	#6	#7	#6	#8	#9	#6	#8	#5	#7

3 Pre-Registered Confirmatory Protocol

To enforce research integrity and eliminate publication bias, the entire test evaluation was pre-registered in `PREREGISTRATION.md` (version 2.0.0, commit tag `prereg-v2`) and cryptographically sealed under Rule R2 prior to data unsealing:

1. **Sealed TEST Partition:** 27 fresh benchmark datasets (Table 1) with zero overlap with development or Paper 1 datasets.
2. **Access Guard:** Access to raw test matrices was programmatically blocked by `guard.py`, raising a fatal `DataPeekingViolation` unless the git tree was clean and tagged at `freeze-v1`.
3. **Frozen Hyperparameters:** All baseline implementations and hyperparameters were frozen (Table 1).
4. **Pre-Registered Decision Tree:**
   - *Branch A (Method Paper):* Triggered if H1 is supported (paired Wilcoxon Holm $p < 0.05$ and Rank #1 on $D/N > 20$).
   - *Branch B (Regime Paper):* Triggered if H1 fails globally, but H2 (HurdleGate recovery) is supported with statistical significance ($\Delta\text{ARI} \geq +0.05, p < 0.05$).
   - *Branch C (Diagnostic Paper):* Fallback if neither H1 nor H2 yields a significant advantage.

4 Empirical Falsification: Continuous Cancer Microarrays

Table 2 presents the results of 5,400 independent model fits across the 15 high-dimensional oncology microarrays ($D/N > 20$).

4.1 Hypothesis H1 Outcome

The primary pre-registered contrast required NullConsensus to achieve a statistically significant improvement over Paper 1's GAC pipeline (icc_v2), CoHiRF, and the negative control gac_random_gating:

- **vs. icc_v2:** $\Delta\text{ARI} = +0.0717$ (95% CI: $[+0.0155, +0.1408]$ strictly > 0), Cohen's $d = +0.56$, $p_{\text{raw}} = 0.0479$, but adjusted $p_{\text{holm}} = 0.1437$ (not significant).

Table 3 Performance recovery on zero-inflated scRNA-seq count cohorts (mean ARI over 10 seeds). HurdleGate decouples Bernoulli dropout point masses from continuous positive support multimodality.

Dataset	ICC-v2 (Paper 1)	NullConsensus	**HurdleGate**	Random Gate	Std. Consensus	Top Baseline
zheng_b_cells	0.041	0.040	**0.788**	0.959	0.992	Ward Hier. (0.993)
zheng_t_cells	-0.027	-0.040	**0.231**	0.341	0.768	Std. Consensus (0.768)
klein_embryonic	0.252	0.274	**0.511**	0.462	0.497	Sparse KM (0.524)
baron_pancreas	0.016	0.042	**0.285**	0.066	0.245	K-Means (0.318)
Mean	0.071	0.079	**0.454**	0.457	0.626	Std. Cons. (0.626)

- **vs. CoHiRF:** $\Delta\text{ARI} = +0.0833$ (95% CI: $[+0.0185, +0.1546]$), Cohen's $d = +0.60$, $p_{\text{raw}} = 0.0479$, $p_{\text{holm}} = 0.1437$ (not significant).
- **vs. gac_random_gating:** $\Delta\text{ARI} = -0.0433$ (95% CI: $[-0.1232, +0.0454]$), $p_{\text{holm}} = 0.2293$.

Crucially, GAC_RANDOM_GATING (Rank #3, Mean ARI = 0.255) outperformed null-screened NULLCONSENSUS (Rank #8, Mean ARI = 0.211). Because $p_{\text{holm}} > 0.05$ and NULLCONSENSUS did not achieve Rank #1, **Hypothesis H1 is formally falsified**.

4.2 Hypothesis H4 Outcome

Hypothesis H4 posited that saliency soft-sampling reduces cross-seed variance by at least 30% ($\overline{R}_{\text{upper}} < 0.70$). The omnibus geometric mean variance ratio was $\overline{R} = 0.6640$, but the 95% bootstrap confidence interval was $[0.2419, 2.0056]$. Because the upper bound exceeds 0.70, **Hypothesis H4 is falsified**.

4.3 Mechanistic Explanation of Falsification

Why did uniform random subspace gating match or exceed statistical null screening on continuous microarrays? In small-sample cohorts ($N < 100$), biological driver genes often display subtle, multi-gene shift patterns rather than pronounced unimodal departures. When evaluated against a finite-sample permutation null ($B = 100$), these true drivers fall below the conservative statistical threshold ($z_d < 1.0$), causing NULLGATE to exclude weakly expressed driver signals. In contrast, uniform random gating samples broadly across the entire transcriptome, retaining diffuse manifold signals that dense consensus averaging successfully aggregates.

5 The Zero-Inflation Discovery: HurdleGate Recovery

While feature gating failed to provide a unique advantage on continuous microarrays, single-cell RNA-sequencing matrices revealed the exact opposite phenomenon.

Table 3 summarizes the 520 model evaluations across the 4 zero-inflated scRNA-seq cohorts ($N = 976$–$1{,}147$, $D = 2{,}000$–$13{,}714$).

5.1 The Dropout Collapse of Standard Gating

On `zheng_b_cells` ($N = 976, D = 13{,}714, K = 3$), ambient K-Means and Ward achieve near-perfect separation (ARI = 0.989 and 0.993). However, when feature

Table 4 Empirical Adjusted Rand Index across the 8 general multi-class and low-dimensional physical control cohorts.

Dataset	K-Means	Ward Hier.	Spectral	Std. Consensus	Core-Halo	FCAG	Sparse KM	CoHiRF	M3C	ICC-v2 (GAC)	Random Gate	NullConsensus
coil20	0.661	0.661	**0.787**	0.660	0.490	0.508	0.394	0.291	0.661	0.562	0.447	0.406
umist_faces	0.361	0.371	**0.408**	0.370	0.294	0.361	0.269	0.273	0.368	0.235	0.350	0.266
texture16	0.035	0.027	-0.004	0.039	0.033	0.034	0.035	0.009	0.036	0.026	0.020	**0.046**
letter_recognition	0.145	0.147	0.078	0.147	0.130	0.123	0.061	**0.159**	0.148	0.104	0.049	0.138
news20_subset	0.086	0.052	0.000	0.032	**0.094**	0.053	0.010	0.000	0.051	0.000	0.008	0.001
parkinsons	-0.096	0.127	**0.157**	-0.091	0.000	0.045	-0.087	-0.060	-0.091	-0.060	-0.009	-0.048
wdbc	0.664	0.575	**0.761**	0.672	0.000	0.689	0.611	0.058	0.671	0.258	0.436	0.560
ionosphere	0.168	**0.177**	-0.034	0.168	0.000	0.114	0.173	0.015	0.168	0.062	0.159	0.175
Mean	0.253	0.267	**0.269**	0.250	0.130	0.241	0.183	0.093	0.252	0.149	0.183	0.193

screening is applied via marginal bimodality (ICC_V2) or standard unimodal null testing (NullConsensus), performance catastrophically collapses to ARI = 0.041 and 0.040. On `zheng_t_cells`, performance plunges below zero (ARI = −0.027 and −0.040).

This confirms our theoretical prediction: technical dropouts create artificial zero-inflation spikes that mimic extreme bimodality in non-informative background genes, causing standard feature screeners to select completely uninformative noise.

5.2 HurdleGate Confirmation (Hypothesis H2)

By decoupling Bernoulli dropout entropy from the positive continuous support, HurdleGate restores manifold clustering:

- On `zheng_b_cells`: ARI recovers from 0.041 to **0.788** (ΔARI = +**0.7471**).
- On `zheng_t_cells`: ARI recovers from −0.027 to **0.231** (ΔARI = +**0.2580**).
- On `baron_pancreas`: ARI recovers from 0.016 to **0.285** (ΔARI = +**0.2695**).
- On `klein_embryonic`: ARI recovers from 0.252 to **0.511** (ΔARI = +**0.2584**).

Across all 4 cohorts, HurdleGate achieves an omnibus recovery of:

$$\Delta\text{ARI} = +\mathbf{0.3832}, \quad 95\%\ \text{CI: } [+0.3157, +0.4596], \quad p = \mathbf{3.57e-08}. \tag{8}$$

The effect size exceeds the pre-registered +0.05 boundary by more than 7×. Therefore, **Hypothesis H2 is decisively supported, unconditionally triggering Branch B (Regime Paper)**.

6 Omnibus Benchmark & Statistical Ranks

Table 4 presents the results for the remaining 8 general multi-class and physical control datasets.

Across all 27 TEST datasets and 12 universal algorithms, the non-parametric Friedman rank test reveals highly significant global performance differences:

$$\chi^2 = \mathbf{52.39}, \quad p = \mathbf{2.31e-07}. \tag{9}$$

6.1 Pairwise Contrasts Across 27 Datasets

Evaluating NullConsensus against prior art across the full benchmark confirms its robustness:

- **vs. icc_v2 (Paper 1):** $\Delta\text{ARI} = +\mathbf{0.0542}$ (95% CI: $[+0.0137, +0.0989]$), $p = \mathbf{1.1200e-02}$ (statistically significant).
- **vs. CoHiRF:** $\Delta\text{ARI} = +\mathbf{0.0689}$ (95% CI: $[+0.0103, +0.1299]$), $p = \mathbf{5.5203e-02}$.

While NullConsensus successfully resolves the instability of Paper 1's GAC pipeline across general datasets, classical dense consensus (std_consensus, Rank #2) and Ward's hierarchical clustering (Rank #1) remain exceptionally competitive general-purpose baselines.

7 Taxonomy of High-Dimensional Gating Regimes

Integrating these empirical discoveries allows us to construct a rigorous practitioner's decision framework:

1. **Continuous Microarrays ($N < 100, D > 1{,}000$, Gaussian-like):** *Avoid marginal statistical gating.* Uniform random subspace sampling or full-feature graph spectral clustering should be preferred, as permutation null filters act over-conservatively on subtle multi-gene driver shifts.
2. **Zero-Inflated Count Manifolds (scRNA-seq, Dropouts $> 50\%$):** *Marginal gating without hurdle correction is catastrophic.* Standard bimodality metrics collapse to noise (ARI ≈ 0). Practitioners must deploy two-part hurdle gating (HurdleGate) to isolate positive-support expression from technical zero point masses.
3. **General Multi-Class & Vision ($K \geq 10$):** Graph spectral clustering and dense ensemble accumulation provide superior topological stability over greedy landmark approximations.

8 Conclusion

Through an unalterable pre-registered confirmatory benchmark spanning 6,880 evaluations across 27 sealed datasets, this study establishes the empirical boundaries of feature-gated consensus clustering. We falsify the assumption that unsupervised null gating provides a universal advantage over uniform random subspaces in continuous high-dimensional regimes, while uncovering and solving the technical dropout collapse in single-cell transcriptomics via HurdleGate. All raw benchmarks, reproducible scripts, and cryptographic hashes are permanently archived in the project repository.

Data and Code Availability. All 27 standardized benchmark datasets, download scripts, and SHA256 checksums are documented in `data/test_manifest.json`. Source code and reproduction pipelines are available under the MIT License at https://github.com/nullconsensus/nullconsensus.

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